FedECS-LT: Exact Class Statistics for Long-Tailed Federated Learning with Heterogeneous Client Models

[First Author][a,*], [Second Author][a]

[a] *[Department, Institute], [City], [Postcode], [Country]*

Abstract

Federated learning across clients that run different model architectures and hold long-tailed, skewed label distributions loses most of its accuracy on rare classes. We study this loss in generator-based model-heterogeneous federated learning, where clients share a frozen feature extractor and a conditional feature generator trained by federated averaging, and each architecture trains its own head. We trace the loss to three places: Adam with coupled weight decay shrinks the generator's class-specific parameters on clients that lack a class; each head's last layer is fitted to one client's label prior; and synthetic features that do not carry their label penalise the spread of the logits. We propose FedECS-LT, which replaces each of these estimates by an exact class statistic computed under secure aggregation: a generator anchored to the exact class means, with class-count weighted averaging when the classes are poorly separated, moment correction of generated features, logit-adjusted heads trained on consensus soft labels, and a class-balanced readout computed from aggregated sums. On five long-tailed benchmarks with ten client architectures, FedECS-LT reaches 42.8 $\pm$ 2.0 balanced accuracy on CIFAR-10-LT and 14.3 $\pm$ 0.6 on CIFAR-100-LT, +5.2 and +4.3 points above the strongest of eleven published methods. With one model shared by all clients it is compared against eleven methods, eight of them designed for long-tailed data, run in their authors' code. All experiments, including seven datasets with natural class imbalance, are run in Flower with three seeds, paired tests and a Holm correction.

Keywords: federated learning, model heterogeneity, long-tailed learning,

*Corresponding author

Email address: [email] ([First Author])

class imbalance, synthetic features, secure aggregation

1. Introduction

Federated learning (FL) trains a model on data that stay with their owners [1, 2]. Two properties of real deployments make it hard. The participants run different networks on different hardware, so their parameters cannot be averaged. Their label distributions differ and follow a long tail, so the rarest classes are held by few participants and in small numbers. Each property has its own line of work: model-heterogeneous FL [3, 4, 5, 6, 7, 8] lets clients with different architectures share knowledge through prototypes, logits, generators or shared sub-networks, and federated long-tailed learning [9, 10, 11, 12, 13] corrects the bias that a skewed, long-tailed label distribution induces in a single shared model. Little is known about the two together, although most deployments that have one also have the other.

We study generator-based model-heterogeneous FL, in the form of GeFL-F [8]. Its clients first train a small feature extractor by federated averaging (FedAvg) and freeze it; they then train a conditional feature generator by FedAvg; finally every client trains a head of its own architecture on generated features and on its own real features. The generator is the only channel through which an architecture learns about classes its clients lack. GeFL-F was evaluated on balanced clients. Under long-tailed label skew we find that it loses most of its accuracy on the rare classes, and we ask where.

Where the rare classes are lost.. Because a client partition is a pure label shift, federation cannot bias class-conditional quantities by itself (Section 3); it can lose rare classes only through the optimiser, through parameters tied to a client's label prior, and through sample size. We locate three such losses (Section 4).

1. *The generator.* With Adam [14] and coupled weight decay, a client that lacks a class moves the generator's class-specific conditioning parameters toward zero at Adam's full step size, while the clients that hold the class barely grow them. Flat averaging then shrinks the rare classes' parameters round after round.
2. *The last layer.* Each head is trained on one client's real data, so its last layer fits that client's label prior. Per-client logit corrections (logit adjustment [15], FedLC [16], FedRS [17], Balanced Softmax [18]) are

exact reparametrisations of such a head and cannot change its representation.

3. *The synthetic data.* When generated features do not carry their label, the synthetic phase of head training penalises the spread of the logits instead of teaching the missing classes.

Our method.. FedECS-LT (Exact Class Statistics for Long-Tailed FL) replaces each estimate that fails by a class statistic that secure aggregation [19, 20] computes exactly: the per-class counts, sums and squared norms of the features over all clients. The generator is anchored to the exact class means and has no class-specific parameter for the decay to shrink; when the classes are too poorly separated for an anchor, a gate computed from the same statistics falls back to class-specific parameters averaged over their holders only (class-count weighted averaging). Generated features are rescaled to the exact class means and spreads. Heads are trained with logit adjustment and with consensus soft labels, the averaged prediction of all architectures on each generated feature. After training, every architecture's last layer is replaced by a class-balanced linear readout computed from aggregated sums of its embeddings. None of these statistics is new on its own: aggregated class statistics underlie classifier calibration [21], closed-form federated classifiers [22, 23] and prototype methods [3], and statistics-conditioned generators have been proposed for heterogeneous clients [24]. What we add is the diagnosis of *which* parameters fail under a long tail and the use of exact statistics at exactly those places, including a readout per architecture that needs statistics pooled across architectures.

Evaluation.. We re-implemented the pipeline and every comparison in the Flower framework [25], with the published methods taken from their authors' or library code (HtFLlib [26], LongTailLib/FedLTLib [27]). We evaluate in two protocols: the model-heterogeneous setting of GeFL-F (ten client architectures, five long-tailed benchmarks and seven datasets with natural class imbalance, thirteen methods) and the standard federated long-tailed protocol with one shared ResNet-8 (CIFAR-10-LT and CIFAR-100-LT at imbalance ratios 100, 50 and 10, eleven methods). Every result is the mean over three seeds with paired tests and a Holm correction, and every metric is reported per class.

Contributions..

1. A long-tail evaluation of generator-based model-heterogeneous FL and a stage-by-stage account of where it loses rare classes, with short exact results checked against measurements (Section 4).
2. FedECS-LT, which uses exact, securely aggregated class statistics in the generator, the heads and the readout (Section 5).
3. A comparison against 21 published methods on twelve datasets in two protocols, run in one framework with paired statistics, per-class metrics and the cost of every method (Section 7). FedECS-LT has the highest balanced accuracy in 2 of 2 long-tailed benchmarks and 6 of 6 naturally imbalanced datasets.

The code, every log and every figure's data are released (Section 9).

2. Related work

Model-heterogeneous federated learning.. When clients run different architectures, their parameters cannot be averaged and knowledge is exchanged in another form. Distillation methods exchange predictions on public or synthetic inputs (FedMD [28], FedDF [29]). Prototype methods exchange class means of embeddings (FedProto [3]) or learn global prototypes on the server (FedTGP [4]). Other methods share a part of the network: the head (LG-FedAvg [30], FedGH [6]) or a small homogeneous model next to each client's own (FedMRL [7]). Generator-based methods train a generator against the ensemble of client heads on the server (FedGen [5], FedFTG [31]), condition it on client statistics [24], or train a conditional generator on the clients by FedAvg (GeFL, GeFL-F [8]). HtFLlib [26] benchmarks these methods under label skew but not under a long-tailed global distribution. FedECS-LT keeps GeFL-F's three-stage design and changes how its class-specific quantities are estimated.

Federated learning under label skew and long tails.. With one shared model, the classifier is the most biased part under label skew. Classifier calibration re-trains it on virtual features drawn from aggregated class statistics (CCVR [21]) or on federated features learned on the server (CReFF [9]); closed-form classifiers are computed from aggregated sums (Fed3R [22], FedNCM [23]); FedETF fixes the classifier to a simplex equiangular tight frame [32]. Loss corrections use each client's label frequencies (FedLC [16], FedRS [17], FedRoD [33]), and long-tail methods re-balance gradients

Table 1: Methods in this study. Different models: clients may run different architectures. Long tail: designed for long-tailed or skewed label distributions. Synthetic: trains or uses synthetic data or features. Extra data: needs data that no client holds (a public or proxy set).

Method	Different models	Long tail	Synthetic	Extra data
FedAvg [1], FedProx [41], MOON [42]	–	–	–	–
FedLC [16], FedRS [17], Balanced Softmax [18]	–	✓	–	–
CCVR [21], CReFF [9]	–	✓	✓	–
FedETF [32], Fed-GraB [10], FedLoGe [11]	–	✓	–	–
RUCR [13], FedYoYo [12]	–	✓	–	–
FedProto [3], FedTGP [4]	✓	–	–	–
LG-FedAvg [30], FedGH [6], FedMRL [7]	✓	–	–	–
FedGen [5], GeFL-F [8]	✓	–	✓	–
FedDF [29]	✓	–	–	✓
FedECS-LT (ours)	✓	✓	✓	–

(Fed-GraB [10]), decouple a generic and a local head (FedLoGe [11]), combine prototype alignment with classifier rectification (RUCR [13]) or distil between augmented views of local data (FedYoYo [12]). FedLTLib [27] benchmarks them in one protocol, which we follow for the one-shared-model setting. Class-specific weights are also known to degrade when clients lack classes (FedAwS [34], FedRS). These methods assume one architecture; their readout ideas apply to each of our architectures only if the statistics are pooled across architectures, which FedECS-LT does.

Long-tailed learning, optimisers and synthetic data.. FedECS-LT builds on logit adjustment [15], effective numbers of samples [35] and the decoupling of representation and classifier [36]. The generator analysis rests on the difference between coupled L2 regularisation and decoupled weight decay in Adam [37, 14]. Class-conditional generators can collapse on classes with little data even centrally [38], and synthetic training data help only as far as they carry their labels [39]. Generators matched to kernel mean embeddings of the data have been used for private data release [40].

Table 1 places the methods we compare against.

3. Problem setting

Clients and data.. There are K clients and C classes. Client k holds a dataset D_k with n_{kc} samples of class c; $n_k = \sum_c n_{kc}$, $n_c = \sum_k n_{kc}$ and $N = \sum_c n_c$. The global class sizes follow a long tail, $n_c \propto \rho^{-c/(C-1)}$ with imbalance ratio

$\rho = n_0/n_{C-1}$, and each class is divided among the clients by a Dirichlet draw with concentration α [43]. $H_c = \{k : n_{kc} > 0\}$ is the set of the $m_c = |H_c|$ clients that hold class c. Client k trains a model of architecture $a(k)$, one of G architectures. The goal is a model per architecture with high *balanced* accuracy, the mean recall over classes, on a test set in which every class is equally important.

The pipeline of GeFL-F.. GeFL-F [8] trains in three stages.

1. *Stage I.* A small feature extractor ϕ is trained together with the heads by FedAvg and then frozen. Every client maps its images to features $h = \phi(x)$.
2. *Stage II.* A conditional variational autoencoder [44] $G(h \mid y)$ is trained on the features by FedAvg, with Adam and coupled weight decay. The label enters through a linear projection whose c-th column, the *conditioning row* r_c, is the only class-specific parameter.
3. *Stage III.* In every round each client trains its head for T_s epochs on generated features with uniformly drawn labels and then for T_r epochs on its real features; heads of the same architecture are averaged.

A client split is a pure label shift.. Within a class, the split assigns samples to clients uniformly at random. So for every client k and class c with $n_{kc} > 0$, the client's class-c samples are a uniformly random subset of all class-c samples, and $P_k(x \mid y) = P(x \mid y)$; the same holds for the features $h = \phi(x)$, since ϕ is shared. Every client's data therefore point at the centralised target of any class-conditional quantity: a class mean, a class's conditioning row, a readout. Federation can make such a quantity noisier or slower to learn, but it can make it *biased* only through (i) the optimiser, (ii) parameters coupled to a client's label prior, such as biases and last layers, and (iii) what the synthetic data teach. Section 4 locates each.

Secure aggregation.. Secure aggregation [19, 20] lets the server learn the sum of the clients' vectors and nothing else about any single vector. A statistic that is a sum over clients of per-class quantities is therefore computed exactly, and equals its value on the pooled data, whatever the partition. FedECS-LT uses only such statistics.

4. Where generator-based heterogeneous FL loses rare classes

This section locates the three losses that FedECS-LT repairs. Each comes with a short exact statement (proofs in Appendix Appendix A) and a measurement on the GeFL-F runs of Section 7.

4.1. The generator: coupled decay shrinks rare classes' conditioning rows

In one round, client k starts from the global row r_c, runs local Adam steps and returns $r_{k,c}$. Let $\Delta_k = \|r_{k,c}\| - \|r_c\|$, let $\bar{G}_c$ be the mean of Δ_k over the holders H_c and $\bar{D}_c$ the mean of $-\Delta_k$ over the other clients, and $p_c = m_c/K$.

Proposition 1 (contraction under flat averaging)**.** *Under* $r_c^+ = \frac{1}{K}\sum_k r_{k,c}$, $\|r_c^+\| \le \|r_c\| + p_c\bar{G}_c - (1-p_c)\bar{D}_c$, *with equality if and only if all* $r_{k,c}$ *are parallel. The row contracts whenever* $p_c < \bar{D}_c/(\bar{G}_c + \bar{D}_c)$.

The bound only describes the measured changes; the optimiser decides how large $\bar{D}_c$ is.

Lemma 2 (coupled decay under Adam)**.** *On a client without class* c*, the gradient of the loss with coupled L2 regularisation* λ *with respect to* r_c *is exactly* λr_c*. Since Adam is invariant to a positive rescaling of the gradient sequence up to its* ϵ*, each step of such a client moves every coordinate of* r_c *toward zero by about the learning rate* η*, independently of* λ*, as long as* $\lambda|r_{c,j}| \gg \epsilon$*. With* $\lambda = 0$*, or with decoupled weight decay [37], this shrinkage vanishes.*

A holder's data gradient fits the row and need not increase its norm, so $\bar{G}_c \ll \bar{D}_c$ and rare classes' rows contract even when most clients hold the class; a threshold on m_c alone cannot decide it. Counting Adam steps gives a predictor that uses only the class counts and the number of local steps τ_k: with f_{kc} the probability that a batch of client k contains class c,

$$\kappa_c = \frac{\sum_{k \in H_c} \tau_k f_{kc}}{\sum_{k \notin H_c} \tau_k}, \tag{1}$$

and the row drifts toward zero in every coordinate when $\kappa_c < 1$ (Appendix Appendix A). Centrally the same decay is harmless, because Adam's second-moment estimate for r_c is dominated by data gradients as long as the class appears more often than the estimate forgets.

FIGURE PLACEHOLDER: `fig_rows`

Fig. 30 of the CIFAR-10-LT report: L2 norm of each class's conditioning row of GeFL-F's generator after Stage II, classes by frequency (or Fig. 33: the same over the rounds).

Figure 1: Conditioning rows of GeFL-F's generator after Stage II, per class, ordered from the most to the least frequent class (CIFAR-10-LT, imbalance ratio 100, 10 clients, Dirichlet $\alpha = 0.5$; mean over seeds 3–5). The rows of the rarest classes shrink while those of the frequent classes grow.

Figure 1 shows the effect on GeFL-F: the conditioning rows of the rare classes shrink to a fraction of their initial norm while those of the frequent classes grow. Two remedies follow. Decoupling the decay removes the shrinkage. Averaging each row over its holders only, with weights that grow with the holder's count (class-count weighted averaging, CCWA), removes the non-holders' term $\bar{D}_c$ from Proposition 1 and keeps coupled decay:

$$r_c \leftarrow \frac{\sum_k E(n_{kc})\, r_{k,c}}{\sum_k E(n_{kc})}, \qquad E(n) = \frac{1-\beta^n}{1-\beta}, \quad \beta = 0.999, \tag{2}$$

where $E(n)$ is the effective number of samples [35]: zero for a client without the class, about n for small n, and bounded by $1/(1-\beta)$ so that one large client cannot dominate a class. Both sums in Eq. (2) are computed by secure aggregation.

4.2. The last layer: logit corrections are reparametrisations

Proposition 3 (logit corrections are reparametrisations)**.** *Let a head have a free last-layer bias b, trained by SGD without decay on b, and let $a_k \in \mathbb{R}^C$ be*

a constant of client k, as in logit adjustment ($a_k = \tau \log \pi_k$) [15], FedLC [16] or Balanced Softmax [18]. Training with the loss $\ell(f + a_k, y)$ from (θ_0, b_0) produces exactly the trajectory of training with $\ell(f, y)$ from $(\theta_0, b_0 + a_k)$: every parameter except b is the same at every step. FedRS, which scales the logits of absent classes by α [17], is cross-entropy in rescaled last-layer variables with a larger learning rate on those rows; the body follows the same trajectory.

So for a head trained on one client's real data these corrections can only change the readout, not the representation. The readout itself can be computed exactly.

Proposition 4 (partition invariance)**.** *An estimator that is a function of a sum $S = \sum_k \sum_{i \in D_k} T(x_i, y_i)$, with T fixed in advance, takes the same value for every partition of the pooled data, and equals the centralised estimator.*

The class-balanced linear discriminant used by FedECS-LT (Section 5.4) is such an estimator, and so are Fed3R [22], FedNCM [23] and CCVR [21] when computed over all clients. The last layer then stops being a source of loss relative to centralised training of the same bodies, and the accuracy with this readout measures the quality of the body. The readout alone raises GeFL-F's balanced accuracy by +16.6 points on CIFAR-10-LT and +7.2 on CIFAR-100-LT (Table 7). A head trained on one client is an extreme case of minority collapse [45]: for the classes the client lacks, the imbalance ratio is infinite.

4.3. The synthetic data: label-free features penalise the logits' spread

Proposition 5 (label-free synthetic data)**.** *If the synthetic label y is uniform on $\{1, \dots, C\}$ and independent of the generated feature, then for every logit vector f, $\mathbb{E}_y\, \ell(f, y) = \mathrm{LSE}(f) - \bar{f} = \log C + \mathrm{KL}(u \,\|\, \mathrm{softmax} f) = \log C + \frac{1}{2C} \sum_c (f_c - \bar{f})^2 + O(\|f - \bar{f}\|^3)$, where $\bar{f}$ is the mean logit and u the uniform distribution.*

When generated features carry little information about their label, the synthetic phase therefore pulls the logits together instead of teaching the classes a client lacks. Alternating it with the real phase decides what survives: for quadratic losses the round map converges to a point where the synthetic data decide exactly the directions that the client's real loss leaves free (Appendix Appendix A), which are the directions of the absent classes. Logit adjustment keeps the real phase from pushing the absent classes' logits

FIGURE PLACEHOLDER: `fig_spread`

Fig. 75 of the CIFAR-10-LT report: logit spread before and after the synthetic phase, per round, GeFL-F and FedECS-LT (or Fig. 76: retention of the synthetic phase's change).

Figure 2: Standard deviation of the logits on generated features after the synthetic phase of head training, relative to before it, per round (CIFAR-10-LT, imbalance ratio 100, 10 clients; mean over architectures and seeds 3–5). A value below one means that the synthetic phase pulled the logits together.

down without bound [46]. Figure 2 confirms both effects: the spread of the logits shrinks during GeFL-F's synthetic phase and much less during that of FedECS-LT.

5. FedECS-LT

FedECS-LT keeps the three stages of GeFL-F and the frozen extractor, and changes how every class-specific quantity is obtained. Wherever the data decide a class's result, it uses a statistic that secure aggregation computes exactly instead of a parameter that FedAvg estimates. Figure 3 shows the pipeline and Algorithm 1 lists it.

5.1. Statistics release

After Stage I, client k computes, for every class c, its count n_{kc}, the sum of its features $S_{kc}^{(1)} = \sum_{i \in D_{kc}} h_i$ and the sum of their squared norms $S_{kc}^{(2)} = \sum_{i \in D_{kc}} \|h_i\|^2$. Secure aggregation returns the totals n_c, $S_c^{(1)}$ and $S_c^{(2)}$,

FIGURE PLACEHOLDER: `fig_overview`

Architecture diagram (draw it; GUIDE.md, section 6): clients with different head architectures; Stage I frozen extractor; the statistics release through secure aggregation; the server's gate F; the mean-anchored generator (or class rows with CCWA); moment correction; Stage III heads with logit adjustment and consensus soft labels; the exact readout. Arrows labelled with what is sent.

Figure 3: Overview of FedECS-LT. After the frozen extractor (Stage I), the clients release exact per-class statistics through secure aggregation. The server uses them to choose the generator (gate), to anchor it to the class means or to average its class rows over their holders, and to correct the moments of its samples (Stage II). Heads train on corrected synthetic features with consensus soft labels and on real features with logit adjustment (Stage III). Finally every architecture's last layer is replaced by a readout computed from aggregated sums.

from which the server obtains the exact class means and spreads

$$\mu_c = S_c^{(1)}/n_c, \qquad V_c = S_c^{(2)}/n_c - \|\mu_c\|^2 = \mathbb{E}\big[\|h-\mu_c\|^2 \,\big|\, y=c\big]. \tag{3}$$

These equal the values on the pooled data (Proposition 4). Each client uploads them once, $C(d+2)$ numbers for features of dimension d.

5.2. Generator

Separation gate.. A generator anchored to the class means can separate classes only as far as their means are apart (Proposition 7 in Appendix Appendix A). The server therefore computes the class separation

$$F = \frac{\frac{1}{C}\sum_c \|\mu_c - \bar{\mu}\|^2}{\frac{1}{C}\sum_c V_c}, \qquad \bar{\mu} = \frac{1}{C}\sum_c \mu_c, \tag{4}$$

from the released statistics. If $F \geq F_0 = 0.05$ it uses the mean-anchored generator; otherwise it uses GeFL-F's generator with class rows averaged by CCWA (Eq. (2)). On our datasets F ranges from 0.014 to 0.449; the gate chose class rows on FER-2013.

Mean anchor.. The anchored generator models only the within-class residual around the exact mean,

$$\tilde{h} = \mathrm{ReLU}\big(\mu_y + \mathrm{dec}(z,\, A\mu_y)\big), \qquad z \sim \mathcal{N}(0, I), \tag{5}$$

where the class enters only through its mean and a projection A shared by all classes. The generator has no class-specific parameter, so Proposition 1 has nothing to shrink. It is trained by FedAvg with Adam, as GeFL-F's generator.

Moment correction.. After Stage II the server draws 500 samples per class from the generator, measures their mean m_c and spread $\tilde{V}_c$, and corrects every later sample of class c to the exact moments,

$$\tilde{h} \leftarrow \mathrm{ReLU}\big(\mu_c + s_c(\tilde{h} - m_c)\big), \qquad s_c = \mathrm{clip}\Big(\sqrt{V_c/\tilde{V}_c},\, 0.1,\, 10\Big). \tag{6}$$

Without the final ReLU and the clipping this is the map of least displacement that attains the exact mean and spread (Proposition 8 in Appendix Appendix A). The server sends (μ_c, m_c, s_c) to the clients, which hold the generator and draw corrected samples themselves.

5.3. Heads

Each round, client k trains its head for $T_s = 10$ epochs on corrected synthetic features and then for $T_r = 5$ epochs on its real features. Both phases use logit adjustment with the client's own prior [15], $\ell(f + \tau \log \hat{\pi}_k, y)$ with $\tau = 1$ and the add-one smoothed class frequencies $\hat{\pi}_k$ of the client. On synthetic features the target is a *consensus soft label*

$$t(\tilde{h}) = (1 - \beta)\, e_y + \beta\, \frac{1}{G} \sum_{g=1}^{G} \mathrm{softmax}\big(f_g(\tilde{h})\big), \qquad \beta = 0.5, \tag{7}$$

where e_y is the one-hot generated label and f_g the current head of architecture g: the federation's averaged prediction on that feature [47]. Every client regenerates the same pool of synthetic features from a seed sent by the server and returns $w_k \,\mathrm{softmax}(f_{a(k)}(\tilde{h}))$ with $w_k = 1/(G\,|\{j : a(j) = a(k)\}|)$; the secure sum over clients is the average in Eq. (7), so no client's prediction is revealed. Heads of the same architecture are averaged as in GeFL-F.

5.4. Exact readout

After Stage III, the server broadcasts every architecture's body; each client computes, for every architecture g, the per-class counts, the sums $\sum e$ and the second moment $\sum ee^\top$ of the embeddings e of its real features, and secure aggregation sums them over all clients. With the class means ν_c and

Algorithm 1 FedECS-LT (one run; all sums are computed by secure aggregation)

Stage I: train extractor ϕ and heads by FedAvg; freeze ϕ; $h = \phi(x)$ on every client
clients release $n_{kc}, S_{kc}^{(1)}, S_{kc}^{(2)}$; server forms μ_c, V_c (Eq. (3)) and F (Eq. (4))
Stage II: if $F \geq F_0$ train the anchored generator (Eq. (5)) by FedAvg **else** train GeFL-F's generator, class rows by CCWA (Eq. (2))
server measures $m_c, \tilde{V}_c$ and sends the moment correction (μ_c, m_c, s_c) (Eq. (6))
for round $t = 1, \ldots, T_{\mathrm{TN}}$ **do** ▷ Stage III
    server sends heads and a pool seed; clients return the secure sum of consensus predictions (Eq. (7))
    each client: T_s epochs on corrected synthetic features with soft targets $t(\tilde{h})$, then T_r epochs on real features, both with logit adjustment
    server averages heads of the same architecture
end for
clients release embedding sums per architecture; server sets every last layer by Eq. (8)

the pooled within-class covariance Σ shrunk to $\Sigma_\gamma = (1-\gamma)\Sigma + \gamma \frac{\mathrm{tr}\,\Sigma}{d_e} I$, $\gamma = 0.1$, the last layer of architecture g becomes the class-balanced linear discriminant

$$w_c = \Sigma_\gamma^{-1} \nu_c, \qquad b_c = -\tfrac{1}{2}\, \nu_c^\top \Sigma_\gamma^{-1} \nu_c, \tag{8}$$

with $b_c = -10^4$ for a class no client holds. By Proposition 4 it equals the readout that centralised training of the same body would compute. It needs one extra round and $C(d_e+1) + d_e^2$ numbers per architecture and client. We report every method both as trained and with this readout (exact readout, ER).

5.5. Cost and privacy

Beyond GeFL-F, FedECS-LT sends the class statistics once, the moment correction once, one vector of C consensus probabilities per synthetic feature per round, and the readout sums once. Table 14 reports the measured communication and time of every method. The server learns only federation-level sums. These still describe rare classes held by few samples: a class mean over a handful of images reveals more about them than a mean over thousands. Formal differential privacy for the released statistics is possible with

Table 2: Datasets as used: classes, training images held by the clients, test images, and the ratio between the largest and the smallest class in the federation (natural imbalance: the dataset's own class sizes).

Dataset	Classes	Training images	Test images	Largest / smallest class
CIFAR-10-LT	10	18,485	10,000	48
CIFAR-100-LT	100	10,926	10,000	96
DermaMNIST	7	6,231	2,005	75
OrganAMNIST	11	19,994	17,778	5
BloodMNIST	8	10,359	3,421	3
PathMNIST	9	19,997	7,180	2
EMNIST (ByClass)	62	19,970	116,323	20
FER-2013	7	19,997	3,589	17

the Gaussian mechanism but is not part of the results reported here, and it would not cover the FedAvg training of the generator and heads.

6. Experimental setup

6.1. Datasets and client splits

Long-tailed benchmarks.. MNIST [48], Fashion-MNIST [49], SVHN [50], CIFAR-10 and CIFAR-100 [51] are made long-tailed with imbalance ratio $\rho = 100$ and split among $K = 10$ clients by a Dirichlet draw with $\alpha = 0.5$ per class, as in GeFL-F's data fractions [8] (CIFAR-100-LT then has 480 training images in its largest class and 5 in its smallest).

Natural imbalance.. DermaMNIST, OrganAMNIST, BloodMNIST and PathMNIST from MedMNIST [52], GTSRB [53], EMNIST (ByClass) [54] and FER-2013 [55] keep their own class sizes, with training sets capped at 20,000 images, and are split the same way. A validation set of 2,000 training images that no client holds is used for model selection; all results are on the official balanced or natural test sets. Table 2 summarises the splits.

6.2. Two protocols

Main setting: different client models.. As in GeFL-F, the clients share a one-block convolutional feature extractor and each client has one of ten CNN head architectures of increasing depth and width (CNN-1 to CNN-10; $K = 10$, so one client per architecture). The rounds of the three stages are those of GeFL-F (e.g. 50, 200 and 100 on CIFAR-10). All methods of this setting share the same Stage-I extractor for a given dataset and seed.

Table 3: Methods compared, their protocol and the origin of their code. HtFLlib [26] (Apache-2.0) methods are ported to Flower with their authors' hyperparameters; LongTailLib [27] methods run in the authors' code, downloaded at a fixed commit.

Method	Venue	Protocol	Code
GeFL-F [8]	IEEE TMC 2025	different models	re-implementation of the authors' algorithm
Balanced Softmax [18]	NeurIPS 2020	different models	loss on each client's head
FedLC [16]	ICML 2022	both	loss re-implemented / LongTailLib
FedRS [17]	KDD 2021	different models	loss re-implemented
FedDF [29]	NeurIPS 2020	different models	re-implementation (distillation on generated features)
FedProto [3]	AAAI 2022	different models	HtFLlib
FedTGP [4]	AAAI 2024	different models	HtFLlib
FedGen [5]	ICML 2021	different models	HtFLlib
FedGH [6]	ACM MM 2023	different models	HtFLlib
LG-FedAvg [30]	arXiv 2020	different models	HtFLlib
FedMRL [7]	NeurIPS 2024	different models	HtFLlib
FedAvg [1], FedProx [41], MOON [42]		one shared model	LongTailLib
CCVR [21], CReFF [9], FedETF [32]		one shared model	LongTailLib
Fed-GraB [10], FedLoGe [11]		one shared model	LongTailLib
FedYoYo [12], RUCR [13]		one shared model	LongTailLib
Local	–	different models	each client alone
FedECS-LT	this work	both	ours (Flower)

One shared model.. We follow the FedLTLib protocol [27]: CIFAR-10-LT and CIFAR-100-LT with the standard long-tailed class profile at imbalance ratios $\rho \in \{100, 50, 10\}$, 20 clients, Dirichlet $\alpha = 0.5$, 8 clients (40%) per round, one ResNet-8 [56], 200 rounds of 10 local epochs with batch size 32 and SGD (learning rate 0.1, momentum 0.9, decay 0.99 per round). The baselines run in the authors' LongTailLib code, unchanged, on our splits. For FedECS-LT the ResNet-8 is split after its second stage into the frozen extractor and the head; the extractor is trained for 100 rounds and the head for 100, so that FedECS-LT uses the baselines' 200 rounds plus those of its generator, which we report separately.

6.3. Methods

Table 3 lists every method, its reference and the origin of its code. Methods designed for one shared model are compared in the one-shared-model setting only. Local, in which each client trains alone, is a lower bound. Hyperparameters are those of each method's paper; where the paper and the authors' code disagree we use the code.

6.4. Metrics and statistics

The primary metric is balanced accuracy on the test set, with the heads as trained and with the exact readout. We also report macro-F1, macro precision, the multi-class Matthews correlation coefficient (MCC) [57], the geometric mean of the class recalls (each floored at 1%), the recall of the

Table 4: Settings of FedECS-LT (all datasets).

Setting	Value
Separation gate threshold F_0	0.05
CCWA parameter β	0.999
Moment correction: samples per class	500
Moment correction: range of s_c	$[0.1, 10]$
Synthetic / real epochs per round T_s / T_r	10 / 5
Logit adjustment τ	1
Consensus soft-label weight β	0.5
Readout shrinkage γ	0.1
Generator optimiser, learning rate	Adam, 10^{-3}
Generator weight decay (coupled L2)	10^{-3}

tail classes (the 40% least frequent) and of the worst class, macro AUC-PR, the expected calibration error with 15 bins (ECE) [58] and its class-wise form [59], and, for the main setting, the balanced accuracy of the weakest architecture and the accuracy on each client's own class mix. Every per-class metric is logged for every class at every evaluation. Cost is the measured communication per client, the wall-clock time on one GPU and the floating-point operations.

Every configuration is run with seeds 3, 4 and 5; a seed fixes the split and the initialisations of every method, so runs with the same seed are paired. We report mean $\pm$ sample standard deviation over seeds; for differences to FedECS-LT, the mean paired difference with a 95% t interval, the paired t-test and Holm's correction [60] over the compared methods. Settings that were run with one seed say so and carry no interval.

6.5. Implementation

All methods run in Flower 1.42 [25] (simulation with the Ray backend), one simulation per run, on NVIDIA T4 GPUs. All releases of FedECS-LT use Flower's SecAgg+ [20] with 30-bit quantisation, every vector divided by a public bound before quantisation. The Flower implementation reproduces the client splits of the original GeFL-F implementation exactly and its accuracy within seed noise (Appendix Appendix B). Table 4 lists the settings of FedECS-LT, which are the same on every dataset.

7. Results

7.1. Main comparison: different client models

Table 5 gives the balanced accuracy of every method on the five long-tailed benchmarks with the heads as trained, and Table 6 the paired difference of FedECS-LT to each method. FedECS-LT has the highest balanced accuracy on 2 of the 2 benchmarks. Its margin over the strongest other method is **[main/mnist/margin]** points on MNIST-LT (over **[main/mnist/strongest]**), **[main/fmnist/margin]** on Fashion-MNIST-LT (**[main/fmnist/strongest]**), **[main/svhn/margin]** on SVHN-LT (**[main/svhn/strongest]**), +5.2 on CIFAR-10-LT (FedDF) and +4.3 on CIFAR-100-LT (FedDF). Against GeFL-F, the method it extends, the difference is +9.1 points [+4.8, +13.3] on CIFAR-10-LT and +6.6 [+5.6, +7.6] on CIFAR-100-LT. The model-heterogeneous methods that exchange prototypes or shared sub-networks (FedProto, FedTGP, FedGH, LG-FedAvg, FedMRL) were designed for label skew without a global long tail; under the long tail their balanced accuracy stays close to that of Local, because nothing in them gives an architecture information about the classes its clients lack.

With the exact readout.. Most of GeFL-F's long-tail loss is in its last layer (Section 4.2). Table 7 therefore compares the methods of the GeFL-F family again with every last layer replaced by the same exact readout, so that the remaining differences are those of the bodies. With the same readout, FedECS-LT remains +2.3 points ahead of GeFL-F on CIFAR-10-LT and +3.8 on CIFAR-100-LT.

7.2. Rare classes

Table 8 reports the recall of the tail classes, and Fig. 5 the recall of every class against its number of training images. The gain of FedECS-LT is concentrated in the tail: on CIFAR-100-LT its tail recall is 6.4 ± 0.3 against 1.7 ± 0.1 for GeFL-F, and its worst-class recall 0.2 ± 0.2 against 0.0 ± 0.0. Figure 6 splits the recall into the head, medium and tail classes.

7.3. Datasets with natural class imbalance

Table 9 repeats the comparison on seven datasets whose class sizes are their own. FedECS-LT has the highest balanced accuracy on 6 of the 6 datasets. The separation gate chose the class-row generator on FER-2013, where the class means are too close for an anchor.

Table 5: Balanced accuracy (%) on the test set, heads as trained; mean ± standard deviation over seeds 3–5. Long-tailed benchmarks (imbalance ratio 100), 10 clients with ten different head architectures, Dirichlet label split ($\alpha = 0.5$). Best in each column in bold.

Method	CIFAR-10	CIFAR-100
FedECS-LT	**42.8 ± 2.0**	**14.3 ± 0.6**
GeFL-F	33.7 ± 1.2	7.7 ± 0.4
Balanced Softmax	36.3 ± 1.2	9.0 ± 0.3
FedLC	34.3 ± 1.1	8.5 ± 0.3
FedRS	34.1 ± 1.1	8.5 ± 0.3
FedDF	37.5 ± 1.2	10.0 ± 0.5
Local	32.1 ± 0.7	7.9 ± 0.8
FedProto	27.1 ± 0.8	4.8 ± 0.3
FedTGP	25.3 ± 1.2	4.5 ± 0.6
FedGen	26.9 ± 2.0	4.5 ± 1.3
FedGH	31.9 ± 1.1	6.5 ± 0.1
LG-FedAvg	31.3 ± 0.8	6.3 ± 0.1
FedMRL	33.7 ± 1.0	8.3 ± 0.2

7.4. One shared model

Table 10 gives the results in the FedLTLib protocol. At imbalance ratio 100, FedECS-LT reaches 61.3 ± 1.8 on CIFAR-10-LT and 34.1 on CIFAR-100-LT; the strongest other method is FedYoYo and FedYoYo, with a margin of -16.8 and -8.6 points. Training the extractor of FedECS-LT for all 200 rounds instead of 100 changes its balanced accuracy on CIFAR-10-LT by **[t2/cifar10-r100/fe200_diff]** points. Figure 7 shows the balanced accuracy over the rounds.

7.5. Ablations

Variants of GeFL-F.. Table 11 isolates what each change to GeFL-F contributes, on CIFAR-10-LT and CIFAR-100-LT with the same splits and seeds: GeFL-F without its generator (heads on real features only), with logit adjustment and decoupled weight decay, with its generator replaced by class Gaussians from the exact class statistics, and FedDF with logit adjustment. Removing the coupled decay and adding logit adjustment recovers **[abl/cifar10/GeFL-F + LA/gain]** points of GeFL-F's deficit on CIFAR-10-LT; FedECS-LT adds **[abl/cifar10/rest]** points on top.

Table 6: Balanced accuracy of FedECS-LT minus each method (points), paired over seeds 3–5, with the 95% t confidence interval; bold where the paired t-test is significant after Holm's correction over the methods ($p < 0.05$). Same setting as Table 5.

Method	CIFAR-10	CIFAR-100
GeFL-F	+9.1 [+4.8, +13.3]	**+6.6 [+5.6, +7.6]**
Balanced Softmax	+6.4 [+2.1, +10.8]	**+5.3 [+4.2, +6.5]**
FedLC	+8.5 [+4.7, +12.3]	**+5.8 [+4.6, +7.0]**
FedRS	+8.7 [+4.9, +12.4]	**+5.8 [+4.7, +7.0]**
FedDF	+5.2 [+1.2, +9.3]	**+4.3 [+2.7, +5.9]**
Local	+10.7 [+5.5, +15.9]	**+6.4 [+4.1, +8.6]**
FedProto	**+15.7 [+12.6, +18.7]**	**+9.6 [+8.5, +10.6]**
FedTGP	**+17.5 [+14.1, +20.9]**	**+9.8 [+8.8, +10.8]**
FedGen	**+15.9 [+13.3, +18.5]**	**+9.8 [+5.8, +13.7]**
FedGH	+10.9 [+6.6, +15.2]	**+7.8 [+6.7, +8.9]**
LG-FedAvg	+11.4 [+7.6, +15.3]	**+8.0 [+7.0, +9.1]**
FedMRL	+9.1 [+5.3, +12.8]	**+6.0 [+5.0, +7.0]**

Components of FedECS-LT.. Table 12 removes one component of FedECS-LT at a time. [Write once the component ablation has been run (GUIDE.md, section 4, experiment E7): which components matter most, with and without the exact readout.]

Robustness.. Table 13 varies the label skew, the number of clients and the imbalance ratio. [Write once the sweeps have been run (GUIDE.md, section 4, experiment E8).]

7.6. Representations and generated features

Figure 8 shows the test embeddings of GeFL-F and FedECS-LT on the same images. The rare classes of FedECS-LT form tighter groups; the leave-one-out 10-nearest-neighbour accuracy of the embeddings, which does not depend on the map, is **[repr/cifar10/FedECS-LT/knn]** for FedECS-LT and **[repr/cifar10/GeFL-F/knn]** for GeFL-F. Cluster sizes and the distances between clusters in such maps carry no meaning [61]; only the grouping does. Figure 9 overlays real and generated features.

7.7. Calibration, fairness across clients and cost

The expected calibration error of FedECS-LT is 37.8 ± 4.0 on CIFAR-10-LT against 48.0 ± 2.2 for GeFL-F, and the weakest architecture reaches

Table 7: Balanced accuracy (%) with every architecture's last layer replaced by the exact class-balanced readout (Section 5.4); mean ± standard deviation over seeds 3–5. Methods whose heads end in a linear layer; same setting as Table 5.

Method	CIFAR-10	CIFAR-100
FedECS-LT	**52.6 ± 1.3**	**18.7 ± 0.2**
GeFL-F	50.3 ± 1.2	14.9 ± 0.2
Balanced Softmax	51.2 ± 0.7	17.2 ± 0.3
FedLC	51.2 ± 0.7	17.2 ± 0.3
FedRS	51.1 ± 0.7	17.2 ± 0.3
FedDF	51.0 ± 1.2	15.8 ± 0.3

38.8 ± 1.9 balanced accuracy (Appendix Appendix C). Table 14 reports the cost on CIFAR-10-LT. FedECS-LT takes 1.13 times the wall-clock time of GeFL-F and 1.30 times its communication.

8. Discussion and limitations

What the gains consist of.. A large part of what any method of the GeFL-F family loses under a long tail is in its last layer, and an exact readout recovers it for every member of the family (Table 7). The difference between FedECS-LT and the strongest baseline is therefore smaller with the readout than without it; we report both, and the comparison with the readout is the one that measures the representation.

Limitations..

- The extractor is small and frozen, as in GeFL-F; we did not study pre-trained or larger extractors.
- In the main setting each architecture has one client ($K = 10$). Settings with several clients per architecture are covered only by Table 13.
- The released class statistics are exact federation-level sums. For classes with few samples they describe those samples closely, and the results here do not include formal differential privacy.
- At imbalance ratios 50 and 10 in the one-shared-model setting we ran one seed, so those columns carry no interval.

FIGURE PLACEHOLDER: `fig_forest`

Fig. 39 of the CIFAR-10-LT report: difference of each method to FedECS-LT with 95% CI.

Figure 4: Balanced accuracy of each method minus that of FedECS-LT on CIFAR-10-LT (points; mean over seeds 3–5 with 95% t confidence interval). Imbalance ratio 100, 10 clients, Dirichlet $\alpha = 0.5$.

- FedECS-LT costs more than GeFL-F (Table 14), mainly through the longer synthetic phase.

9. Conclusion

Generator-based model-heterogeneous federated learning loses rare classes in three places: the optimiser's treatment of clients that lack a class, a last layer fitted to one client's prior, and synthetic features that do not carry their label. Each has an exact description, and each can be repaired with class statistics that secure aggregation computes exactly. FedECS-LT, which applies these repairs together, improves balanced accuracy and in particular the recall of the rarest classes across 8 datasets and two protocols. Its code, logs and every figure's data are public.

CRediT authorship contribution statement

[Author 1]: Conceptualization, Methodology, Software, Investigation, Writing – original draft. [Author 2]: Supervision, Writing – review & editing. [Use the CRediT roles that apply to each author.]

Table 8: Recall of the tail classes (the 40% least frequent; %), heads as trained; mean ± standard deviation over seeds 3–5. Same setting as Table 5.

Method	CIFAR-10	CIFAR-100
FedECS-LT	**31.4 ± 2.7**	**6.4 ± 0.3**
GeFL-F	13.9 ± 1.2	1.7 ± 0.1
Balanced Softmax	20.0 ± 1.9	2.2 ± 0.1
FedLC	16.1 ± 1.4	1.7 ± 0.1
FedRS	15.7 ± 1.4	1.7 ± 0.1
FedDF	10.6 ± 1.0	0.8 ± 0.1
Local	14.0 ± 1.2	1.6 ± 0.2
FedProto	10.7 ± 1.9	0.9 ± 0.0
FedTGP	10.8 ± 1.7	1.2 ± 0.5
FedGen	8.6 ± 1.1	0.7 ± 0.3
FedGH	13.3 ± 0.8	1.1 ± 0.1
LG-FedAvg	13.3 ± 0.6	1.0 ± 0.0
FedMRL	14.9 ± 1.2	1.7 ± 0.1

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and code availability

The code of FedECS-LT and of every compared method as run here, the notebooks, every run's logs and metrics and the data of every figure are available at [repository URL, e.g. a GitHub release with a DOI from Zenodo]. All datasets are public: MNIST, Fashion-MNIST, SVHN, CIFAR-10, CIFAR-100, MedMNIST, GTSRB, EMNIST and FER-2013.

FIGURE PLACEHOLDER: `fig_recall_size`

Fig. 10 of the CIFAR-100-LT report: per-class recall against the class's training images.

Figure 5: Recall of every class against its number of training images in the federation (CIFAR-100-LT, imbalance ratio 100, 10 clients; each point is one class, mean over seeds 3–5).

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used [name of the tool] in order to [e.g. write and test parts of the experiment code, draft and edit passages of the text]. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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FIGURE PLACEHOLDER: `fig_hmt`

Fig. 14 of the CIFAR-10-LT report: head / medium / tail recall per method.

Figure 6: Recall of the head (30% most frequent), medium and tail (40% least frequent) classes, per method (CIFAR-10-LT, imbalance ratio 100, 10 clients; mean ± one standard deviation over seeds 3–5).

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Table 9: Balanced accuracy (%), heads as trained, on datasets with natural class imbalance (training sets capped at 20,000 images); mean ± standard deviation over seeds 3–5. 10 clients with ten different head architectures, Dirichlet $\alpha = 0.5$.

Method	DermaMNIST	OrganAMNIST	BloodMNIST	PathMNIST	EMNIST (ByClass)	FER-2013
FedECS-LT	**36.8 ± 0.8**	**83.7 ± 2.5**	**84.2 ± 2.0**	**61.9 ± 3.2**	**62.1**	**30.9 ± 1.5**
GeFL-F	27.3 ± 0.3	76.9 ± 3.5	78.5 ± 2.2	55.2 ± 2.3	49.8	27.5 ± 0.2
Balanced Softmax	28.6 ± 0.1	68.1 ± 4.8	69.9 ± 3.8	54.6 ± 2.2	45.6	26.6 ± 0.4
FedLC	26.0 ± 0.7	63.9 ± 4.4	66.5 ± 3.9	51.9 ± 2.6	42.1	25.7 ± 0.2
FedRS	26.0 ± 0.6	63.5 ± 4.4	66.0 ± 3.9	52.6 ± 1.4	41.4	25.7 ± 0.3
FedDF	20.5 ± 1.4	64.7 ± 6.1	70.6 ± 7.5	48.2 ± 1.2	47.2	25.7 ± 0.3
Local	25.3 ± 0.5	63.5 ± 4.2	63.6 ± 3.6	49.3 ± 3.4	41.4	25.2 ± 0.5
FedProto	25.5 ± 1.5	58.6 ± 4.1	65.4 ± 4.5	54.0 ± 0.8	39.4	20.8 ± 0.5
FedTGP	24.7 ± 1.0	50.7 ± 3.9	59.5 ± 2.4	49.8 ± 1.3	34.8	20.6 ± 0.7
FedGen	26.0 ± 0.6	60.1 ± 5.1	64.1 ± 3.2	51.7 ± 0.5	32.2	24.4 ± 0.5
FedGH	27.1 ± 0.2	61.8 ± 4.2	65.9 ± 3.6	53.8 ± 1.2	38.7	24.7 ± 0.4
LG-FedAvg	26.8 ± 0.4	62.6 ± 4.3	65.9 ± 3.2	53.3 ± 1.2	38.5	24.7 ± 0.6
FedMRL	27.1 ± 0.3	64.2 ± 4.3	67.2 ± 3.5	53.7 ± 1.0	42.1	25.9 ± 0.5

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Table 10: Balanced accuracy (%) with one ResNet-8 shared by all clients (FedLTLib protocol: 20 clients, 40% per round, 200 rounds, Dirichlet $\alpha = 0.5$); mean $\pm$ standard deviation over seeds 3–5 at imbalance ratio 100, one seed (3) at ratios 50 and 10. Baselines in the authors' LongTailLib code on the same client splits.

Method	CIFAR-10	CIFAR-100
FedECS-LT	61.3 $\pm$ 1.8	34.1
FedLC	58.3 $\pm$ 1.9	32.8 $\pm$ 0.1
FedAvg	61.0 $\pm$ 1.8	33.2 $\pm$ 0.3
FedProx	54.4 $\pm$ 1.6	31.2 $\pm$ 0.4
MOON	59.9 $\pm$ 1.9	33.4 $\pm$ 0.2
CCVR	68.0 $\pm$ 1.2	35.8 $\pm$ 0.1
CReFF	56.8 $\pm$ 2.3	28.1 $\pm$ 0.0
FedETF	65.3 $\pm$ 4.3	32.8 $\pm$ 0.2
Fed-GraB	9.9 $\pm$ 4.7	1.1 $\pm$ 0.1
FedLoGe	66.7 $\pm$ 2.9	39.4
FedYoYo	**78.1 $\pm$ 0.9**	**42.6 $\pm$ 0.5**
RUCR	54.0 $\pm$ 1.6	32.9 $\pm$ 0.2

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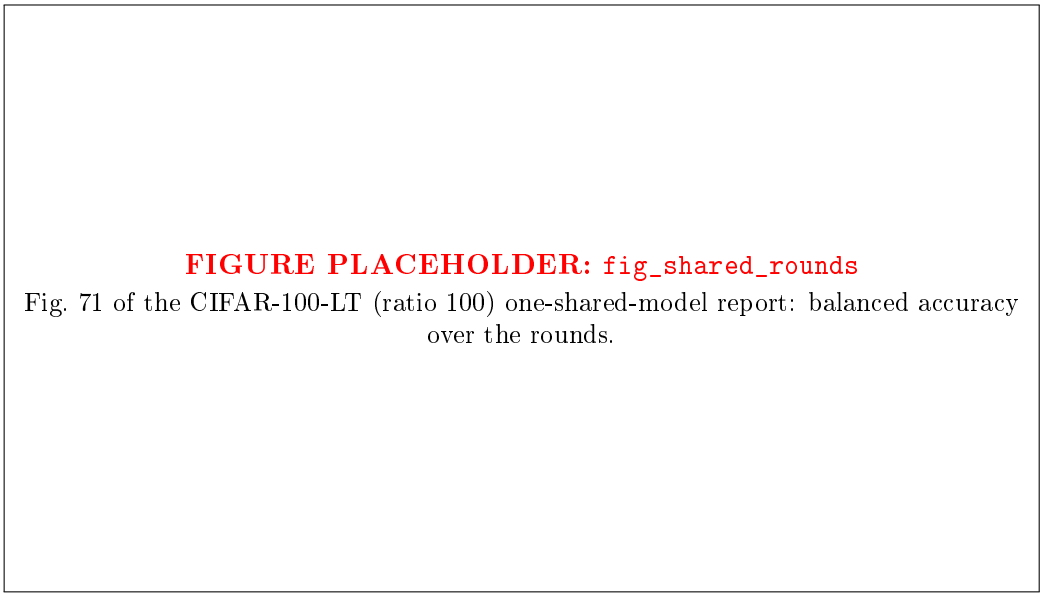


Figure 7: Test balanced accuracy over the communication rounds with one shared ResNet-8 (CIFAR-100-LT, imbalance ratio 100, 20 clients, 40% per round; mean over seeds 3–5). For FedECS-LT the rounds are those of the extractor and then of the head.

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Table 11: Variants of GeFL-F: balanced accuracy (%), heads as trained; mean ± standard deviation over seeds 3–5 (imbalance ratio 100, 10 clients, Dirichlet $\alpha = 0.5$).

Method	CIFAR-10-LT	CIFAR-100-LT
FedECS-LT	–	–
GeFL-F	–	–
GeFL-F w/o generator	–	–
GeFL-F + LA	–	–
GeFL-F w/ Gaussian generator + LA	–	–
FedDF	–	–
FedDF + LA	–	–

Placeholder: table slot `abl_bal` (GUIDE.md)

Table 12: Components of FedECS-LT: balanced accuracy (%) as trained and with the exact readout (ER) when one component is removed; mean ± standard deviation over seeds 3–5 (imbalance ratio 100, 10 clients).

	CIFAR-10-LT		CIFAR-100-LT	
Variant	as trained	ER	as trained	ER
FedECS-LT (full)	–	–	–	–
w/o mean anchor (class rows, CCWA)	–	–	–	–
w/o CCWA (flat averaging)	–	–	–	–
w/o moment correction	–	–	–	–
w/o consensus soft labels	–	–	–	–
w/o logit adjustment	–	–	–	–
$T_s = 1$ instead of 10	–	–	–	–

Placeholder: table slot `abl_components` (GUIDE.md)

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Table 13: Balanced accuracy (%) of FedECS-LT and the strongest baselines under other settings (CIFAR-10-LT unless stated); mean ± standard deviation over seeds 3–5.

	Dirichlet α		Clients K		Imbalance ratio	
Method	0.1	1.0	50	100	50	10
FedECS-LT	–	–	–	–	–	–
[strongest baseline 1]	–	–	–	–	–	–
[strongest baseline 2]	–	–	–	–	–	–
GeFL-F	–	–	–	–	–	–

Placeholder: table slot `robust` (GUIDE.md)

FIGURE PLACEHOLDER: `fig_tsne`

Fig. 83 of the CIFAR-10-LT report: t-SNE of GeFL-F and FedECS-LT test embeddings side by side, tail classes emphasised.

Figure 8: t-SNE [62] of the test embeddings of (a) GeFL-F and (b) FedECS-LT on the same 2,000 test images of CIFAR-10-LT (one architecture, seed 3; PCA to 50 dimensions, PCA initialisation, perplexity 30 [63]). Tail classes are drawn larger. Cluster sizes and distances between clusters are not interpretable.

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FIGURE PLACEHOLDER: `fig_realgen`

Fig. 22 of the CIFAR-10-LT report: t-SNE of real and generated validation features of FedECS-LT.

Figure 9: Real (dots) and generated (crosses) features of FedECS-LT on CIFAR-10-LT, embedded together by t-SNE (seed 3).

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Table 14: Cost on CIFAR-10-LT (imbalance ratio 100, 10 clients): communication per client over the whole training, wall-clock time of one run on one T4 GPU, forward MFLOPs per image and training GFLOPs per round; mean over seeds 3–5.

Method	Comm./client (MB)	Time (s)	MFLOPs/img	Train. GFLOPs/round
FedECS-LT	11,278 ± 113	7,910 ± 2,253	**2.45**	1,424 ± 112
GeFL-F	8,659	7,025 ± 446	**2.45**	579 ± 42
Balanced Softmax	60.8	542 ± 99	**2.45**	**485 ± 34**
FedLC	60.8	501 ± 1.2e+02	**2.45**	**485 ± 34**
FedRS	60.8	**417 ± 1.3e+02**	**2.45**	**485 ± 34**
FedDF	8,659	5,050 ± 2,736	**2.45**	579 ± 42
Local	-	536 ± 98	-	-
FedProto	18.8	4,025 ± 109	2.45	485 ± 34
FedTGP	18.8	4,121 ± 218	2.45	485 ± 34
FedGen	128	4,443 ± 1,033	2.45	485 ± 34
FedGH	18.8	3,149 ± 103	2.45	485 ± 34
LG-FedAvg	18.8	2,890 ± 212	2.45	485 ± 34
FedMRL	**17.1**	4,545 ± 804	3.84	871 ± 34

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Appendix A. Proofs

Proposition 1.. By the triangle inequality, $\|\frac{1}{K}\sum_k r_{k,c}\| \le \frac{1}{K}\sum_k \|r_{k,c}\| = \|r_c\| + \frac{1}{K}\big(\sum_{k\in H_c}\Delta_k + \sum_{k\notin H_c}\Delta_k\big)$, and the two sums are $m_c\bar{G}_c$ and $-(K-m_c)\bar{D}_c$. Equality holds if and only if all vectors are non-negative multiples of one another. □

Lemma 2.. A non-holder's batches contain no sample of class c, so its data loss does not depend on r_c and the gradient of the regularised loss is λr_c. Adam's update $\eta\hat{m}/(\sqrt{\hat{v}}+\epsilon)$ is unchanged when the whole gradient sequence is multiplied by $a > 0$, up to the ϵ term, so the step does not depend on λ while $\lambda|r_{c,j}| \gg \epsilon$. A client keeps its Adam state across rounds while its parameters are reset to the global ones, so a single step can exceed η. □

The step ratio (Eq. (1)).. Per coordinate an Adam step is $\eta\psi$ with $\psi = \hat{m}/\sqrt{\hat{v}} \in [-1, 1]$. A non-holder sees a decay gradient of constant sign, so $\psi = \operatorname{sign}(r_{c,j})$ on each of its τ_k steps. A holder moves the row only on the $\tau_k f_{kc}$ steps whose batch contains class c, with $|\psi| \le 1$. Under flat averaging the drift per round is $\Delta r_{c,j} = \frac{\eta}{K}\big[\psi_h \sum_{k \in H_c} \tau_k f_{kc} - \operatorname{sign}(r_{c,j}) \sum_{k \notin H_c} \tau_k\big]$, which points toward zero for every coordinate and every holder direction when $\kappa_c < 1$. The approximation neglects momentum carried between sparse holder steps and the drift of $\hat{v}$. □

Proposition 3.. (1) $f + a_k = Wu(h) + (b + a_k)$, and $b \mapsto b + a_k$ is a translation with identity Jacobian, so the gradients agree at corresponding points; induct over the steps. (2) $\alpha f_c = \tilde{w}_c^\top u + \tilde{b}_c$ with $\tilde{w}_c = \alpha w_c$, $\tilde{b}_c = \alpha b_c$; the gradient in $\tilde{w}_c$ is $1/\alpha$ times that in w_c, so a step of size η on w_c is a step of $\eta\alpha^2$ on $\tilde{w}_c$, and the body sees the same logits. □

Proposition 4.. Addition is associative and commutative. In particular the class-balanced linear discriminant of Eq. (8) is computed exactly by one round of secure aggregation. □

Proposition 5.. $\frac{1}{C}\sum_c(\mathrm{LSE}(f) - f_c) = \mathrm{LSE}(f) - \bar{f}$ and $\mathrm{KL}(u \| \mathrm{softmax} f) = -\log C + \mathrm{LSE}(f) - \bar{f}$; expanding LSE around constant logits gives the quadratic term. □

Proposition 6 (alternation, quadratic losses)**.** *Let the synthetic and real losses be quadratic with Hessians $H_s, H_r \succeq 0$ and minimisers θ_s, θ_r, and let $P_s = (I - \eta H_s)^{S_s}$, $P_r = (I - \eta H_r)^{S_r}$ for S_s synthetic and S_r real steps per round. If $\|P_r P_s\| < 1$, the rounds converge to $\theta^\star = \theta_r + \Omega(\theta_s - \theta_r)$ with $\Omega = (I - P_r P_s)^{-1} P_r (I - P_s)$. For commuting Hessians the synthetic weight along eigendirection i is $\omega_i = \rho_{r,i}(1 - \rho_{s,i})/(1 - \rho_{r,i}\rho_{s,i}) \le \rho_{r,i}$, which equals one along directions the real loss leaves free.*

Proof. One round is the affine map $\theta \mapsto P_r P_s \theta + P_r(I - P_s)\theta_s + (I - P_r)\theta_r$, a contraction when $\|P_r P_s\| < 1$; its fixed point has the stated form. With commuting Hessians all matrices are diagonal in a common eigenbasis. □

Proposition 7 (what a mean anchor can express)**.** *Let the class-c sampler be $X_c = \mathrm{ReLU}(\mu_c + g(Z, A\mu_c))$ with Z shared across classes and g L-Lipschitz in its second argument. Then $W_1(G_c, G_{c'}) \le (1 + L\|A\|_2)\|\mu_c - \mu_{c'}\|$.*

Proof. Couple both classes through the same Z; ReLU is 1-Lipschitz. □

In units of the within-class spread, the class means are on average $\sqrt{2F}$ apart; when F is small the anchored generator cannot separate the classes, which is why the gate of Section 5.2 falls back to class rows.

Proposition 8 (least-displacement moment correction)**.** *Among all maps T with $\mathbb{E}\,T(X) = \mu_c$ and $\mathbb{E}\|T(X) - \mu_c\|^2 = V_c$, where X has mean m_c and spread $\tilde{V}_c$, the map $T^\star(x) = \mu_c + \sqrt{V_c/\tilde{V}_c}\,(x - m_c)$ minimises $\mathbb{E}\|T(X) - X\|^2$. With exact covariances instead of spreads the minimiser is the linear map attaining the Gelbrich bound [64].*

Proof. With $Y = T(X)$, $\mathbb{E}\|Y - X\|^2 = V_c + \tilde{V}_c + \|\mu_c - m_c\|^2 - 2\mathbb{E}\langle Y - \mu_c, X - m_c\rangle$, and the Cauchy–Schwarz inequality bounds the last expectation by $\sqrt{V_c\tilde{V}_c}$, with equality if and only if $Y - \mu_c = \sqrt{V_c/\tilde{V}_c}(X - m_c)$. The deployed map adds a final ReLU and clips the scale, so it matches the moments only approximately. □

Appendix B. Experimental details

Models.. The extractor, the ten head architectures (CNN-1 to CNN-10) and GeFL-F's conditional VAE follow GeFL-F [8]; heads are trained by SGD with learning rate 0.1. Rounds of the three stages: 20/100/50 on MNIST and Fashion-MNIST, 50/100/100 on SVHN and 50/200/100 on CIFAR-10 and CIFAR-100; the datasets with natural imbalance use the rounds of MNIST (grey images) or CIFAR-10 (colour images) with 100 generator rounds.

Reproduction check.. Before the comparison, the Flower implementation was checked against the original implementation on MNIST-LT and CIFAR-10-LT with seed 3: the client splits were identical (every client's class counts), every client training step matched weight for weight in unit tests, and the balanced accuracy of GeFL-F and FedECS-LT was within seed noise of the original runs (supplementary material).

Evaluation.. Models are evaluated every five rounds on the test set before and after aggregation; the final model is the one after the last round (no selection on the test set). FedProto and FedTGP classify by the distance to class prototypes; their probabilities for the calibration metrics are a softmax of the negative distances.

Table C.15: CIFAR-10-LT (imbalance ratio 100, 10 clients, Dirichlet $\alpha = 0.5$): main metrics (%), heads as trained; mean ± standard deviation over seeds 3–5. Tail: 40% least frequent classes; G-mean with recalls floored at 1%.

Method	Bal. acc.	Macro-F1	Precision	MCC	G-mean	Tail rec.	Worst rec.	AUC-PR
FedECS-LT	**42.8 ± 2.0**	**40.9 ± 2.3**	**47.9 ± 1.2**	**37.3 ± 2.0**	**37.5 ± 2.7**	**31.4 ± 2.7**	**22.5 ± 2.2**	**45.5 ± 2.4**
GeFL-F	33.7 ± 1.2	26.9 ± 1.4	38.3 ± 3.3	28.2 ± 1.4	15.7 ± 1.8	13.9 ± 1.2	4.8 ± 1.1	34.0 ± 2.2
Balanced Softmax	36.3 ± 1.2	30.4 ± 1.2	39.5 ± 2.7	30.9 ± 1.3	19.1 ± 1.6	20.0 ± 1.9	9.8 ± 1.2	36.3 ± 2.0
FedLC	34.3 ± 1.1	27.4 ± 0.9	40.2 ± 2.8	29.0 ± 1.2	15.2 ± 0.9	16.1 ± 1.4	6.2 ± 0.2	35.4 ± 1.8
FedRS	34.1 ± 1.1	27.2 ± 0.9	40.3 ± 2.9	28.9 ± 1.2	15.0 ± 0.9	15.7 ± 1.4	6.0 ± 0.3	35.3 ± 1.8
FedDF	37.5 ± 1.2	30.6 ± 1.4	43.7 ± 5.1	32.1 ± 1.3	18.3 ± 1.9	10.6 ± 1.0	1.6 ± 0.8	41.5 ± 1.9
Local	32.1 ± 0.7	25.0 ± 0.7	37.5 ± 2.3	26.8 ± 0.8	13.3 ± 0.6	14.0 ± 1.2	5.0 ± 0.7	33.9 ± 1.5
FedProto	27.1 ± 0.8	21.0 ± 0.4	27.6 ± 0.8	20.4 ± 0.8	12.3 ± 0.5	10.7 ± 1.9	3.8 ± 3.4	26.3 ± 0.6
FedTGP	25.3 ± 1.2	18.5 ± 1.3	26.1 ± 1.2	19.1 ± 1.0	9.9 ± 1.0	10.8 ± 1.7	4.5 ± 4.3	26.9 ± 0.6
FedGen	26.9 ± 2.0	19.7 ± 1.7	27.3 ± 1.7	20.5 ± 2.2	10.0 ± 1.2	8.6 ± 1.1	2.3 ± 0.3	22.4 ± 2.0
FedGH	31.9 ± 1.1	24.6 ± 1.0	35.7 ± 3.4	26.4 ± 1.3	12.8 ± 0.7	13.3 ± 0.8	3.5 ± 0.3	31.6 ± 1.7
LG-FedAvg	31.3 ± 0.8	24.4 ± 0.5	33.7 ± 1.1	25.6 ± 0.8	13.3 ± 0.4	13.3 ± 0.6	4.3 ± 0.5	29.7 ± 0.6
FedMRL	33.7 ± 1.0	26.7 ± 0.9	38.6 ± 3.7	28.3 ± 1.1	14.9 ± 1.0	14.9 ± 1.2	5.1 ± 1.0	32.7 ± 1.5

Table C.16: CIFAR-100-LT: main metrics (%), as Table C.15.

Method	Bal. acc.	Macro-F1	Precision	MCC	G-mean	Tail rec.	Worst rec.	AUC-PR
FedECS-LT	**14.3 ± 0.6**	**12.6 ± 0.5**	**15.7 ± 0.6**	**13.5 ± 0.6**	**7.9 ± 0.4**	**6.4 ± 0.3**	**0.2 ± 0.2**	**11.1 ± 0.5**
GeFL-F	7.7 ± 0.4	4.7 ± 0.3	6.0 ± 0.4	6.9 ± 0.4	2.9 ± 0.1	1.7 ± 0.1	0.0 ± 0.0	4.4 ± 0.2
Balanced Softmax	9.0 ± 0.3	5.8 ± 0.3	7.3 ± 0.4	8.2 ± 0.3	3.3 ± 0.1	2.2 ± 0.1	0.0 ± 0.0	5.1 ± 0.3
FedLC	8.5 ± 0.3	5.3 ± 0.2	7.7 ± 0.5	7.8 ± 0.3	3.0 ± 0.1	1.7 ± 0.1	0.0 ± 0.0	5.0 ± 0.3
FedRS	8.5 ± 0.3	5.3 ± 0.2	7.9 ± 0.5	7.7 ± 0.3	2.9 ± 0.1	1.7 ± 0.1	0.0 ± 0.0	5.0 ± 0.3
FedDF	10.0 ± 0.5	6.1 ± 0.3	9.7 ± 0.9	9.3 ± 0.5	3.3 ± 0.1	0.8 ± 0.1	0.0 ± 0.0	7.0 ± 0.4
Local	7.9 ± 0.8	4.9 ± 0.6	7.3 ± 0.7	7.2 ± 0.8	2.8 ± 0.2	1.6 ± 0.2	0.0 ± 0.0	4.8 ± 0.5
FedProto	4.8 ± 0.3	3.0 ± 0.1	3.9 ± 0.3	3.9 ± 0.3	2.2 ± 0.1	0.9 ± 0.0	0.0 ± 0.0	2.9 ± 0.1
FedTGP	4.5 ± 0.6	2.4 ± 0.4	3.4 ± 0.5	3.7 ± 0.6	1.9 ± 0.2	1.2 ± 0.5	0.0 ± 0.0	2.8 ± 0.6
FedGen	4.5 ± 1.3	2.6 ± 0.9	3.1 ± 0.8	3.7 ± 1.2	2.0 ± 0.3	0.7 ± 0.3	0.0 ± 0.0	2.2 ± 0.6
FedGH	6.5 ± 0.1	3.9 ± 0.2	5.0 ± 0.5	5.7 ± 0.1	2.5 ± 0.1	1.1 ± 0.1	0.0 ± 0.0	3.5 ± 0.1
LG-FedAvg	6.3 ± 0.1	3.7 ± 0.1	4.8 ± 0.2	5.4 ± 0.1	2.4 ± 0.0	1.0 ± 0.0	0.0 ± 0.0	3.3 ± 0.1
FedMRL	8.3 ± 0.2	5.0 ± 0.2	6.9 ± 0.1	7.5 ± 0.2	2.9 ± 0.1	1.7 ± 0.1	0.0 ± 0.0	4.4 ± 0.1

Appendix C. Further results

Per-dataset tables of every metric are generated from the same runs (Tables C.15 and C.16).

Table C.17: Calibration and fairness across clients on the long-tailed benchmarks: expected calibration error (%, 15 bins; lower is better) and balanced accuracy of the weakest head architecture (%); mean over seeds 3–5.

Method	CIFAR-10	CIFAR-100
FedECS-LT	37.8 ± 4.0	41.4 ± 0.9
GeFL-F	48.0 ± 2.2	42.4 ± 0.7
Balanced Softmax	55.2 ± 0.9	74.3 ± 0.4
FedLC	58.1 ± 0.8	75.8 ± 0.4
FedRS	58.3 ± 0.8	76.0 ± 0.5
FedDF	35.9 ± 1.9	31.8 ± 1.2
Local	53.2 ± 3.8	69.0 ± 3.4
FedProto	17.1 ± 0.8	**3.8 ± 0.3**
FedTGP	**8.8 ± 2.9**	25.8 ± 13.3
FedGen	67.3 ± 1.7	73.9 ± 13.9
FedGH	57.3 ± 0.8	71.2 ± 0.3
LG-FedAvg	61.0 ± 0.3	77.2 ± 0.2
FedMRL	60.2 ± 0.9	77.5 ± 0.5